

What is transform in CSS?

-> The transform property in CSS is used to change the position, size, shape, or rotation of an element without affecting the normal document flow.

-> Using transform, we can:

1. Move elements
2. Rotate elements
3. Resize elements
4. Skew elements
5. Combine multiple transformations

Types of Transform Functions:

-> There are mainly 4 functions:

1. `translate()`
2. `rotate()`
3. `scale()`
4. `skew()`

1. `translate()`:

-> It is used to move an element from its current position.

-> Syntax: `transform: translate(x, y);`

x → horizontal movement

y → vertical movement

-> Individual Translate Functions:

1. `translateX()`: Moves only horizontally.
2. `translateY()`: Moves only vertically.

2. rotate():

-> It is used to rotate the an element.

-> Syntax: transform: rotate(angle);

Positive Value -> clockwise

Negative Value -> anticlockwise

-> By default it will rotate around z-axis.

-> Individual Rotate Functions:

1.rotateX(): Rotate only about x-axis.

2.rotateY(): Rotate only about Y-axis.

3. scale():

-> It is used to increase or decrease element size.

-> Syntax: transform: scale(x, y);

-> Individual Scale Functions:

1. scaleX(): Increases width.

2. scaleY(): Increases height.

-> Important:

1 -> Original size (No change)

>1 -> Increase the size

<1 -> Decrease the size

4. skew():

-> It is used to tilt an element.

-> Syntax: transform: skew(x-angle, y-angle);

-> Individual Skew Functions:

1. skewX(): Tilt an element in X-axis.
2. skewY(): Tilt an element in Y-axis.

Combining Multiple Transforms:

-> We can apply multiple transforms together.

-> Syntax:

transform: translateX(100px) rotate(45deg) scale(1.2);

Transform Origin Property:

-> It is used to change the starting point of transformation.

-> Syntax: transform-origin: x y;

-> Common Values:

Value	Meaning
-------	---------

-----	-----
-------	-------

center	Default
--------	---------

top	Top
-----	-----

bottom	Bottom
--------	--------

left	Left
------	------

right	Right
-------	-------

Perspective in 3D:

-> Perspective is the distance between the viewer(screen) and the element in 3D space.

-> Syntax: perspective: value(px);

-> Small perspective value = object looks more closer

-> Large perspective value = object looks farther away

-> Example:

```
.container {  
    perspective: 500px;  
}
```

-> 500px means the viewer is 500px away from the object.

Important Note:

-> Perspective is usually applied to the parent element, not directly on the child being transformed.

Ideal Range:

Perspective Value	Effect
-----	-----
200px – 500px	Strong 3D effect
600px – 1000px	Natural/realistic effect
1000px+`	Subtle/flat effect